

Theoretical Ecotoxicology

Session: Introduction ecological modelling

University Duisburg-Essen SS 2026

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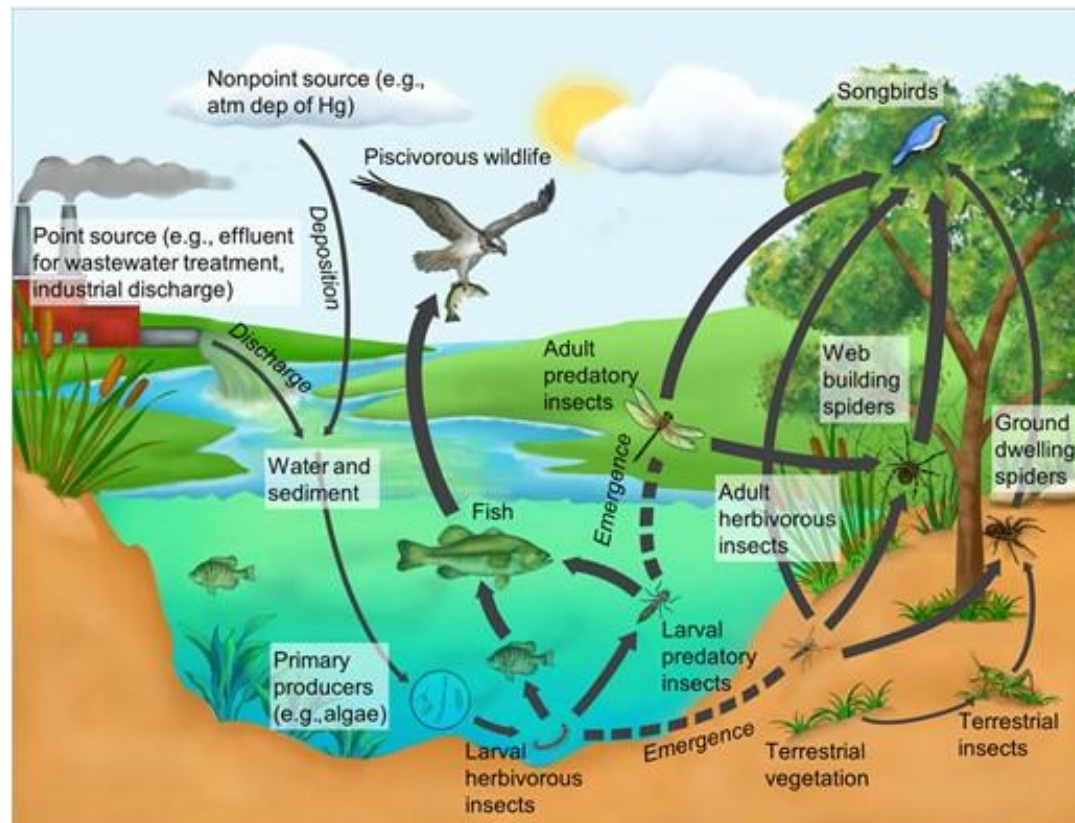
Ecotoxicology, Faculty of Biology, UDE & RC One Health Ruhr

15.07.2026

Aquatic ecosystem and its biodiversity

are under...

- Environmental pressure (e.g., drought and flood)
- Ecological pressure (e.g., predation and invasion)
- Chemical pressure (e.g., pesticides, heavy metals, or PFAS)



Ecological & ecotoxicological modelling

Ecological modelling is:

- A mathematical representation of ecosystems
- Used to understand, simulate, and predict ecosystem dynamics
- Address *natural and anthropogenic pressures* (inc. chemical stress)
- Capture processes across *various levels of biological organisation* and *spatiotemporal scales*

Ecotoxicological models are ecological models built to predict chemical impacts on ecosystems and support risk assessment
(Jørgensen 2009 - *Ecological Modelling: An Introduction*)

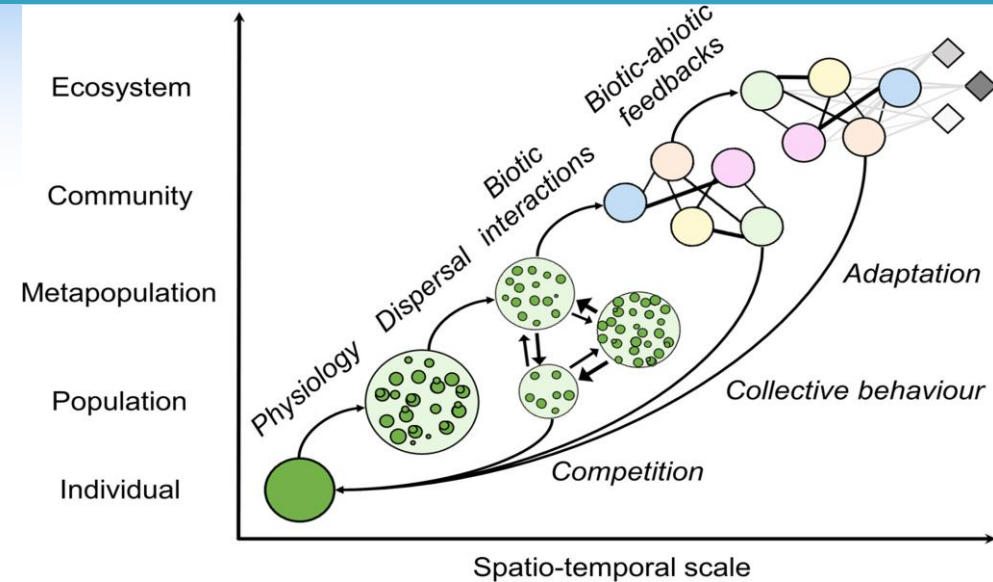
Chapter 2

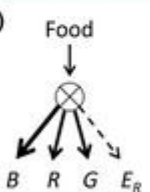
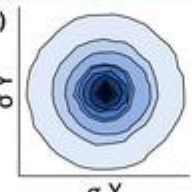
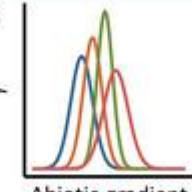
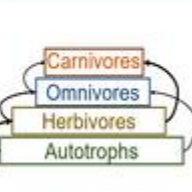
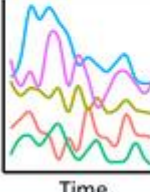
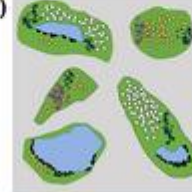
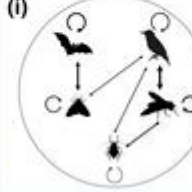
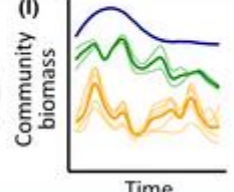
Model types	
2.1 Classification of models	
2.2 Model types – an overview	
2.3 Characteristics of the model types	
2.3.1 Dynamic biogeochemical and bioenergetic models.....	
2.3.2 Steady state biogeochemical models.....	
2.3.3 Population dynamic models.....	
2.3.4 Structurally dynamic models	
2.3.5 Fuzzy models	
2.3.6 Artificial neural networks	
2.3.7 Spatial models.....	
2.3.8 Individual based models and cellular automata	
2.3.9 Ecotoxicological models.....	
2.3.10 Stochastic models	
2.3.11 Hybrid models	
2.4 Applicability of the different model types.	

Ecological modelling

Ecological models:

- Population models
- Metapopulation models
- Metacommunity models
- Individual-based models
- Species interaction models
- Species distribution models
- Spatial models...



	Individual-based model (IBM)	Meta-population model (MPM)	Meta-community model (MCM)	Ecosystem model (EcoM)
Mech./Proc.	(a) Food 	(d) σY 	(g) σX 	(j) Abiotic gradient 
Scale	(b) 	(e) 	(h) 	(k) 
Patterns	(c) Population abundance 	(f) 	(i) 	(l) Community biomass 

Ecological modelling

Today's session focuses:

- Density-dependent population growth model
- Individual-based Model (IBM)
- TKTD – IBM

Population growth model

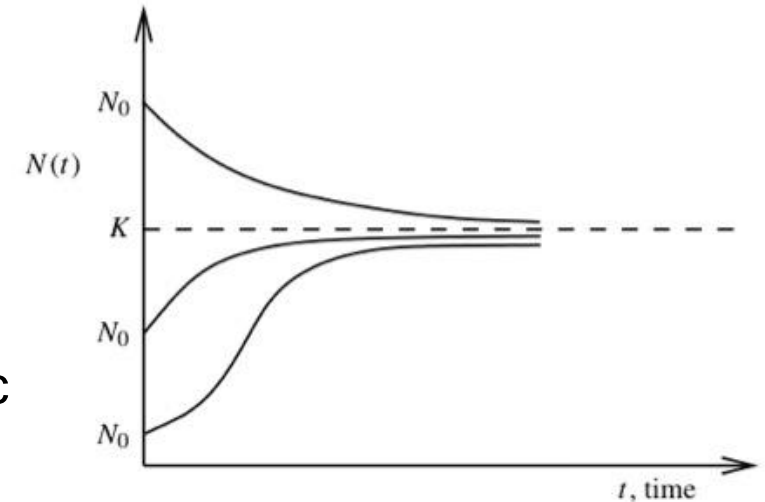
- In classical population ecology, population is the model unit
- Density-dependent population growth model (Verhulst, 1838):

$$\frac{dN}{dt} = rN \left(1 - \frac{N}{K} \right)$$

- N = population
- r = intrinsic (per capita) growth rate
- $K = \frac{1}{\alpha}$: carrying capacity, i.e. maximum population size that environment can support

- Main assumptions:

- All individuals in a population are identical
- Individuals are not modelled explicitly
- Increased density caused intraspecific competition due to limited resources



Murray 2002: 4

Exercise 1 – Population growth model

- Run the population growth model in R

$$\frac{dN}{dt} = rN * (1 - \alpha N)$$

- N = population
- r = intrinsic (per capita) growth rate
- $K = \frac{1}{\alpha}$: carrying capacity, i.e. maximum population size that environment can support

- Manipulate r and α and see how the model reacts

Population to individuals

- In classical population ecology, population is the model unit

$$\frac{dN}{dt} = rN * (1 - \alpha N)$$

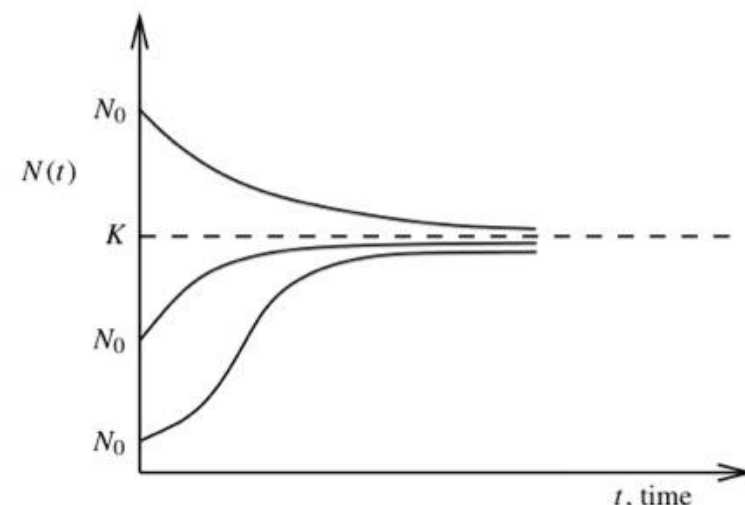
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- r = intrinsic (per capita) growth rate
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- Main assumptions:

- All individuals in a population are identical
- Individuals are not modelled explicitly

- But:

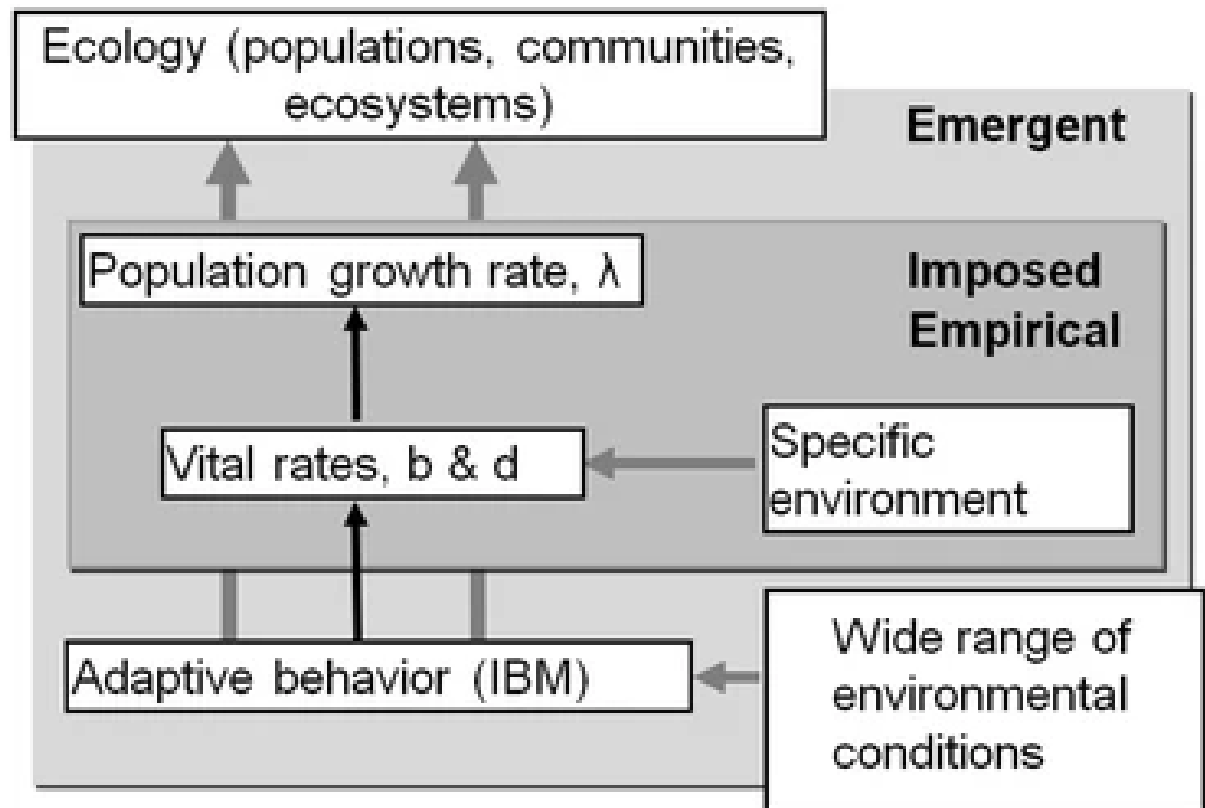
- Individuals may contribute to explain patterns at (meta-) population and (meta-) community levels
- Effects of chemical pressure may strongly differ across levels of biological organization



Basic principle of Individual-Based Model (IBM)

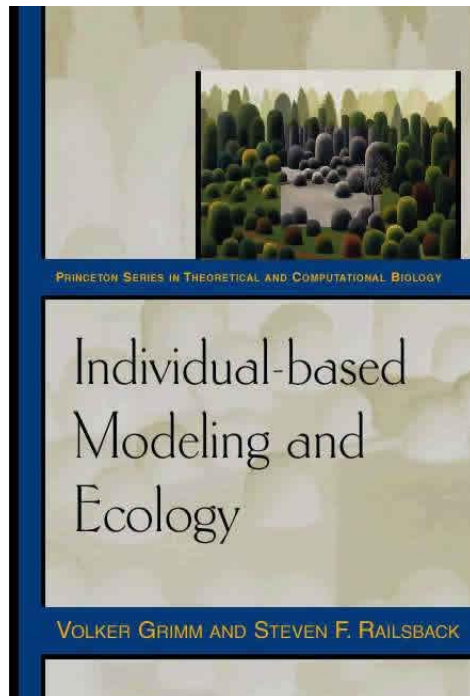
- *Individual organisms* described as autonomous, unique entities with own attributes and behaviors (e.g., age, size, or social rank)
- *System dynamics* emerge from interactions between *individuals* and their environment

IBMs are
bottom-up models



Individual-based Model (IBM) and Agent-based Model (ABMs)

- IBMs and ABMs are used interchangeably:
 - IBMs: common in ecology, focusing on biological traits, life histories, and individual-level interactions
 - ABMs: more common in social sciences and other fields (e.g., economics, demography, geography); agents maybe social groups or grid cells representing a landscape



Grimm V, Railsback SF (2005): *Individual-based Modeling and Ecology*.

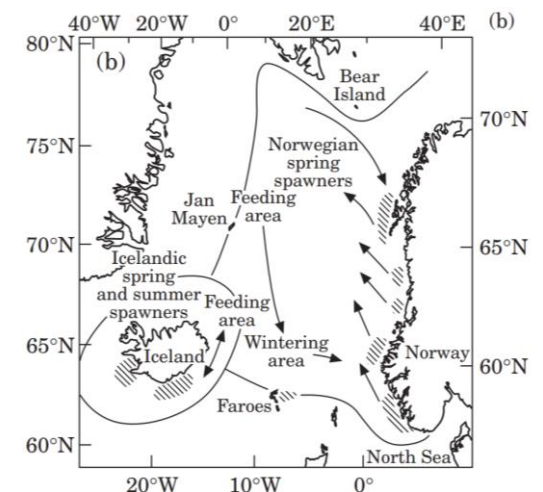
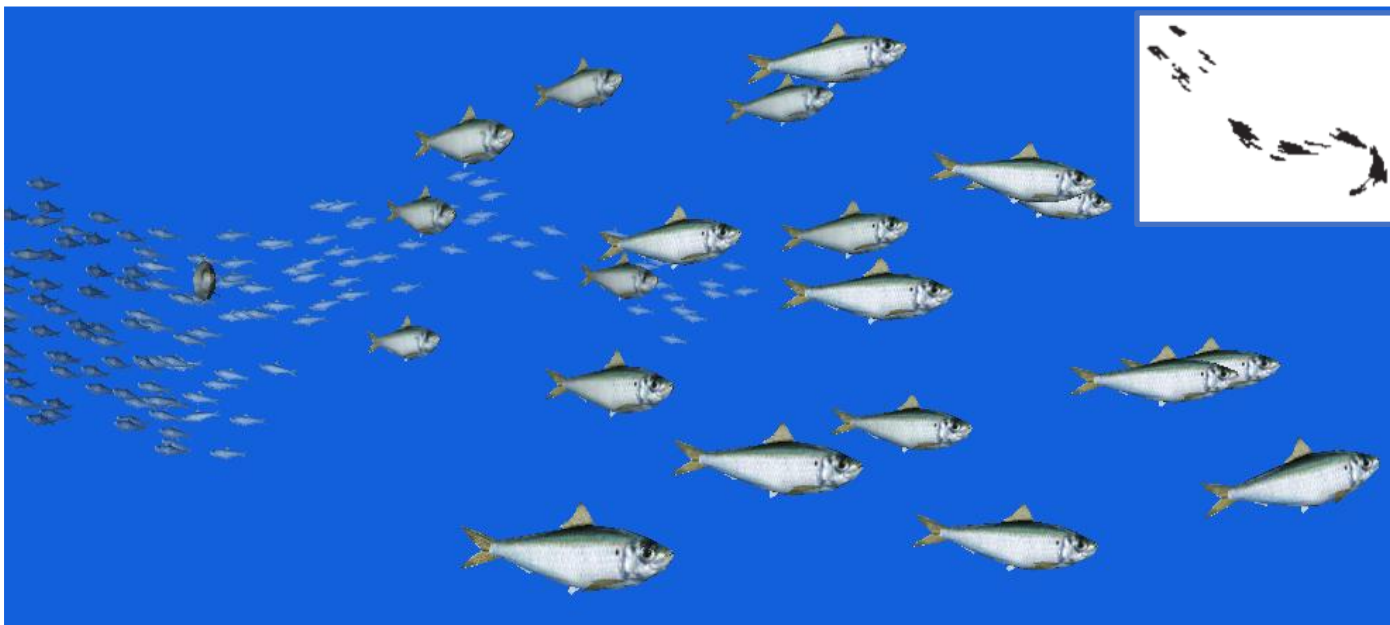


Railsback SF, Grimm V (2009): *Agent-Based and Individual-Based Modeling: Practical Introduction*

Example – IBM-fish

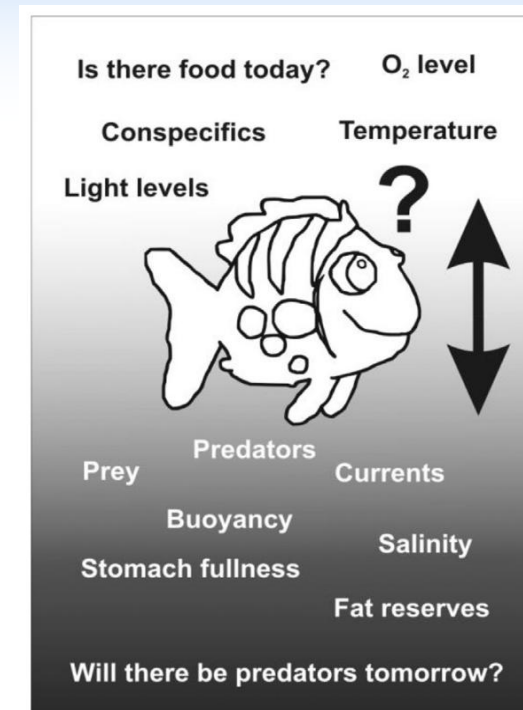
Norwegian spawning herring *Clupea harengus*

- Young herring learn migration routes by schooling with experienced individuals
- Shifts in migration patterns in years of high recruitment
- Large influxes of young recruits can overwhelm the influence of experienced adults, disrupting migratory knowledge transfer
- *Individual* behavior shapes collective migration patterns of *population*



Why to use IBM

- Individuals:
 - Variability in physiology (e.g., even of same age), growth, development, and life history (e.g., change in size and food requirements)
 - Local interactions ($\neq$ global interactions, all individuals interact with all other individuals)
 - Adaptive behavior (i.e., maximize fitness by adapting behavior to their current state and environment)
- Interactions among individuals = spatial location is important
- System dynamics emerge from individual behavior

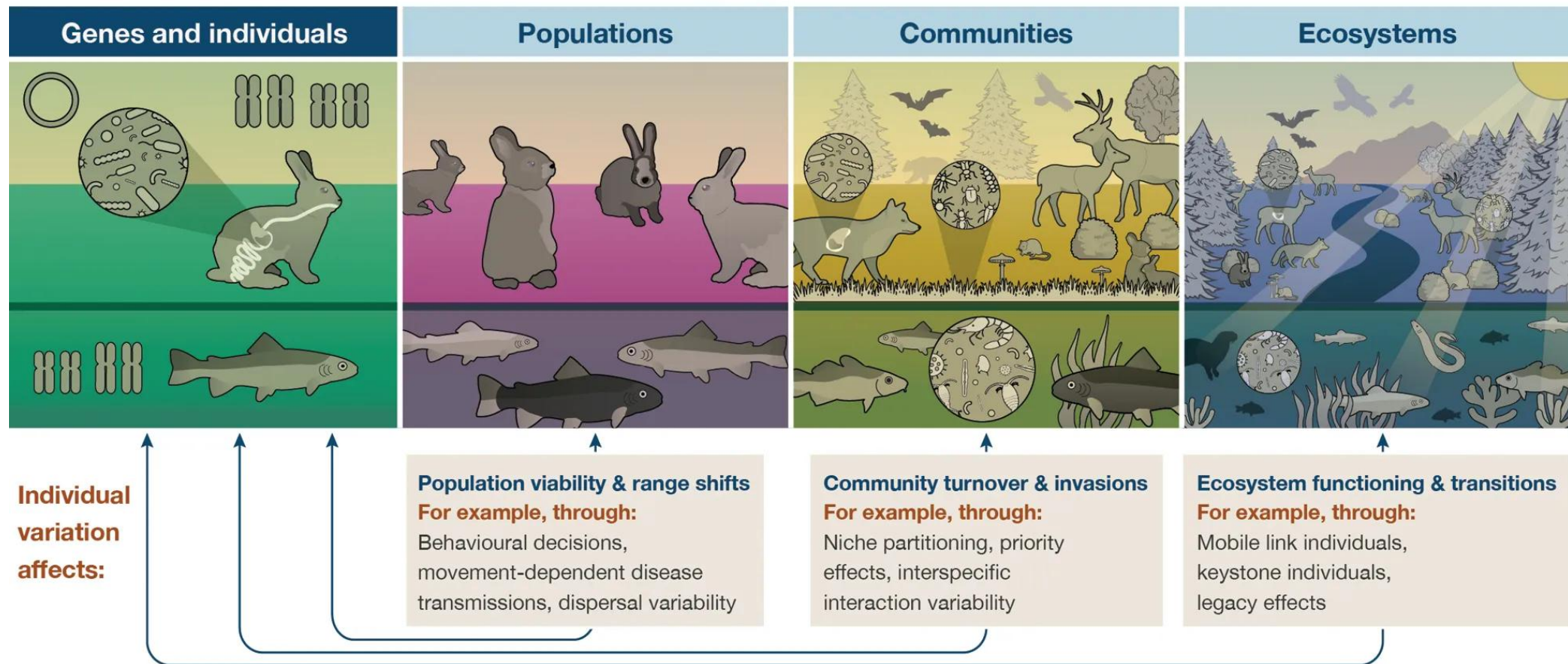


Why to use IBM

Individual variation may become more important
with global environmental change

Global change leads to increased stochasticity, non-equilibrium dynamics and non-stationary systems

→ Individual variation becomes critical to the biodiversity response



Why not to use IBM

- Lack of information: Populations are composed of individuals, but often information is limited to population level
- IBMs are often complex and come at cost:
 - Lack of a conceptual framework and theory → designed *ad hoc*
 - Data hungry, many parameters unknown
 - Uncertainty in model structure
 - Challenging to test
 - Model communication difficult

It depends on the (research) question!

Exercise 2 - IBM model of Daphnia population (Petzoldt & Rinke, 2007)

- **Main research questions:**
 - How individual-level processes scale up to population-level pattern in water flea *Daphnia magna*?
 - Scenarios: How size-selective predation alter population trajectories?
- **Simecol:** a model object includes:
 - *main*: rules executed at each time step
 - *parms*: fixed constants
 - *init*: initial condition
 - *times*: the simulated period
 - *solver*: algorithm iterating the model through time
 - *equations*: model functions



Exercise 2 - IBM model of Daphnia population (Petzoldt & Rinke, 2007)

Workflow:

- *live()*:
 - individual body size increases -> reach maturation (size > SAM)
 - Individual produces a clutch of eggs
- *survive()*: individuals are removed based on maxage (i.e. death age)
- *hatch()*: eggs development, reach eggage, then released as neonates
- New neonates added to the population
- Emerged population: abundance, mean age, mean eggs per individual
- Scenario: mortality by size-selective predation (none-selective/small-preferred/large-preferred)

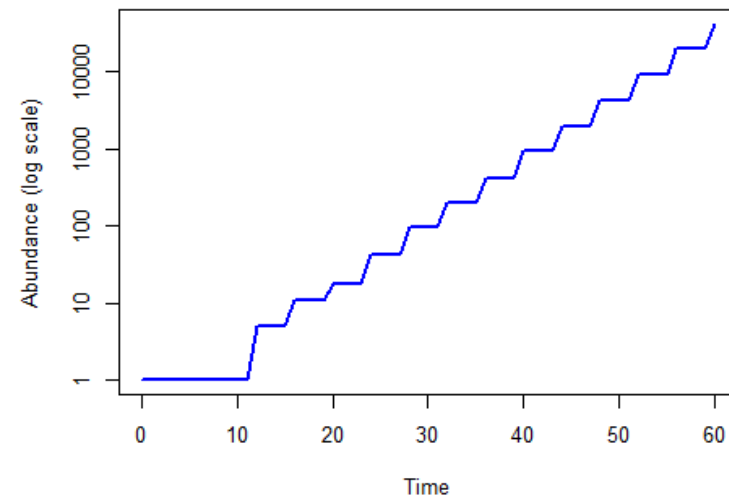
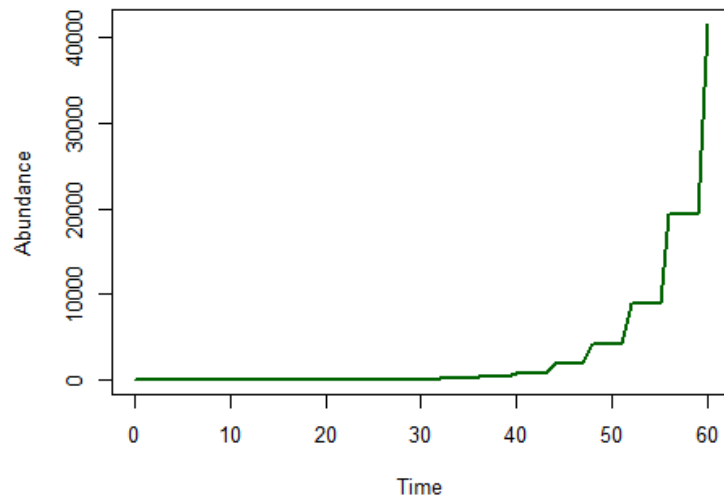
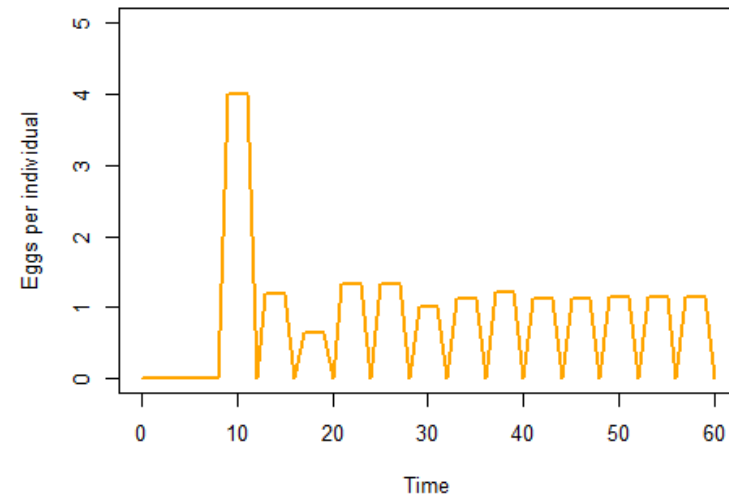
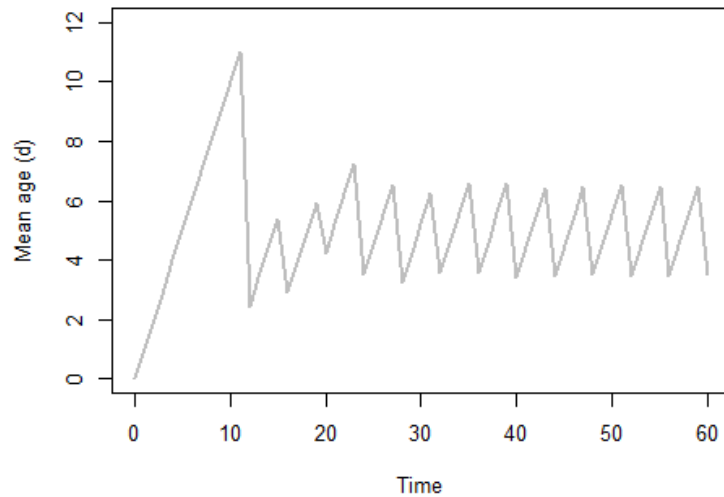
Exercise 2 - IBM model of Daphnia population (Petzoldt & Rinke, 2007)

Main steps:

1. Load library (simecol)
2. Define model object (indbasedModel)
3. Equations (live, survive, hatch)
4. Define user-defined solver (myiteration() + observer())
5. Run simulation (sim())
6. Visualize outcomes (abundance, mean size/age/eggs)
7. Run scenarios

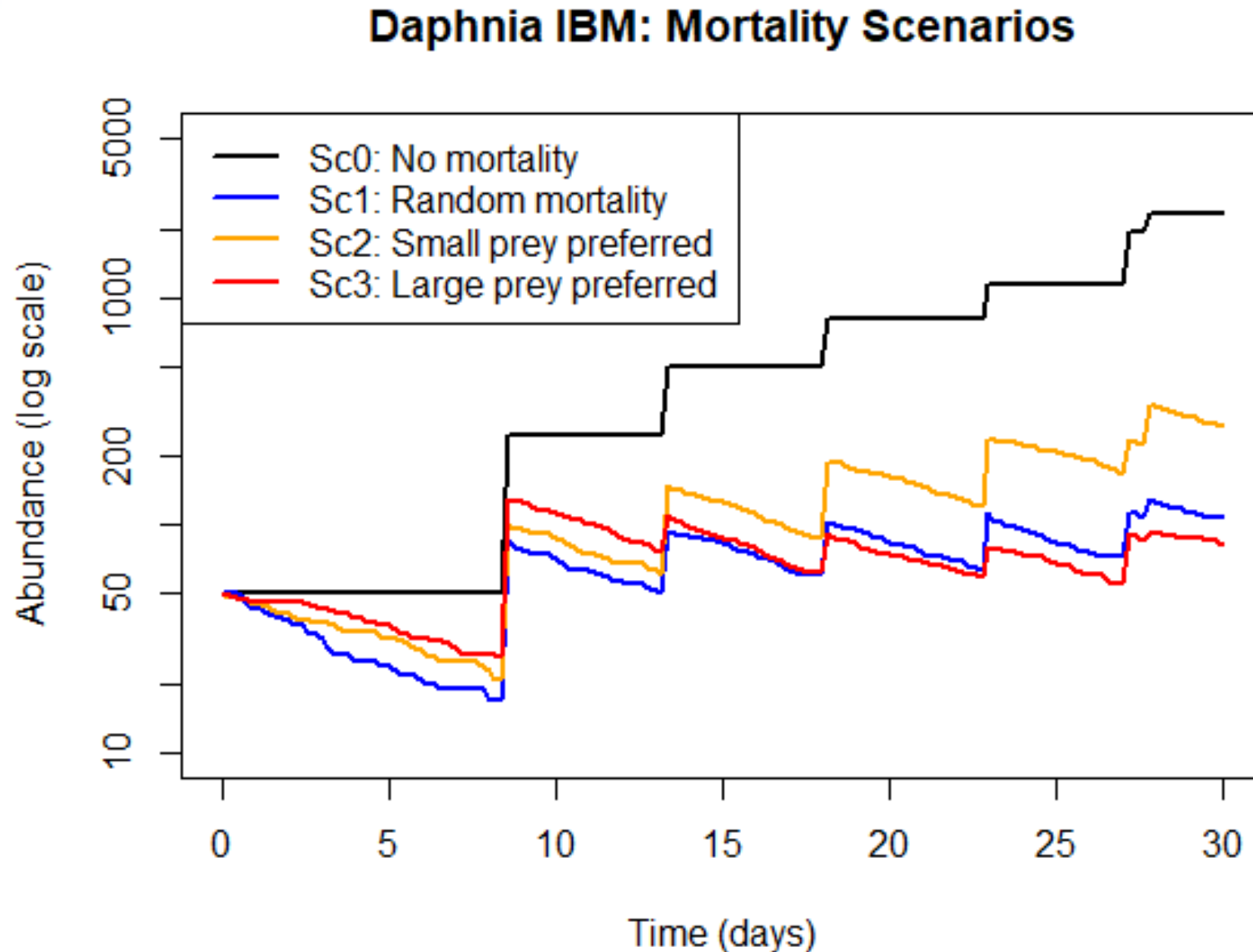
Exercise 2 - IBM model of Daphnia population (Petzoldt & Rinke, 2007)

Model outputs:



Exercise 2 - IBM model of Daphnia population (Petzoldt & Rinke, 2007)

Scenario outputs: SC0 to SC3



Exercise 3 - Coupling toxicity models to an IBM

- **Goal:** predict population-level effects of a chemical from individual-level processes
- **Case study:**
 - *Species:* Daphnia magna population
 - *Chemical:* Copper
 - *Data source:* Vlaeminck et al. (2021) Environ. Toxicol. Chem. 40: 513-527
- **Model components:**
 - *IBM:* same as Exercise 2 (individual growth, egg production, hatching, mortality)
 - *Predation mortality* is replaced by a *mechanistic copper stressor*

Exercise 3 - Coupling toxicity models to an IBM

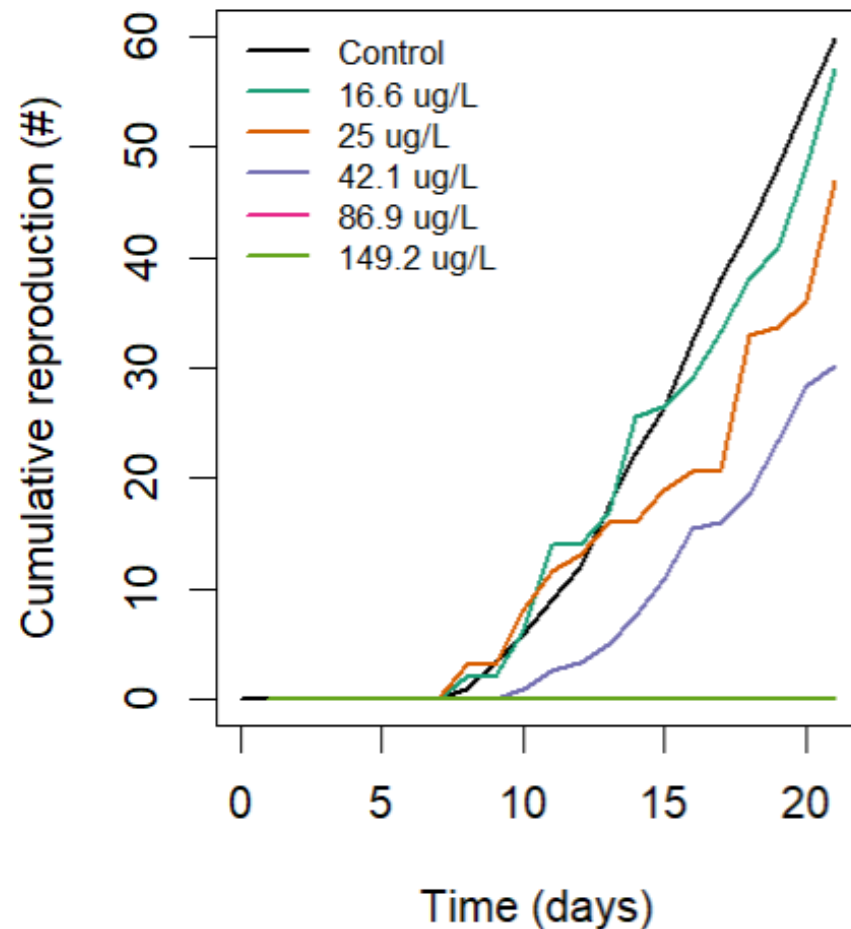
Workflow:

- External copper C_w
- Internal scaled damage D_w ($TKTD$, update rate kD)
- Two effects on each individual Daphnia:
 - $DEBtox$: reduced assimilation -> slower growth and reproduction
 - $GUTS$: stochastic death
- Individual effects
- Emerged population dynamics
- Model validation based on observed data (R^2)

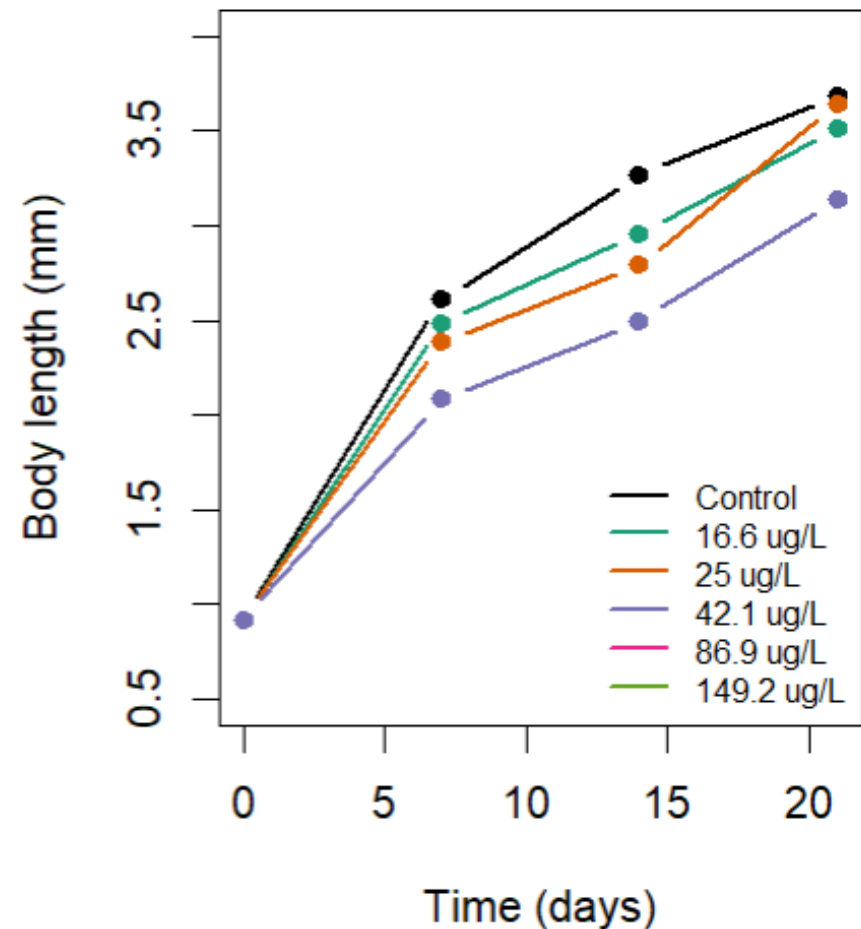
Exercise 3 - Coupling toxicity models to an IBM

DEBtox (sublethal):

Sublethal: reproduction

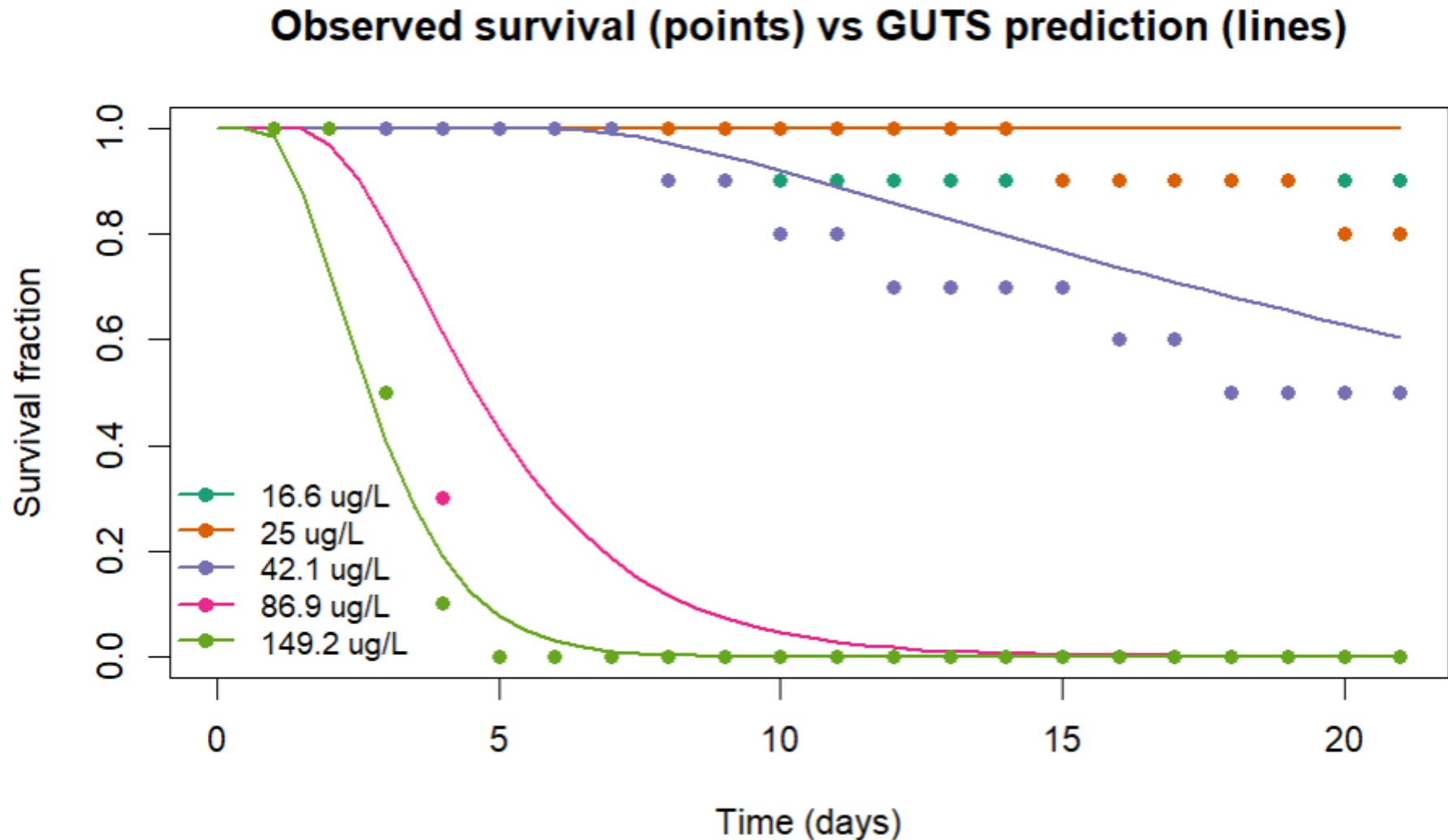


Sublethal: growth



Exercise 3 - Coupling toxicity models to an IBM

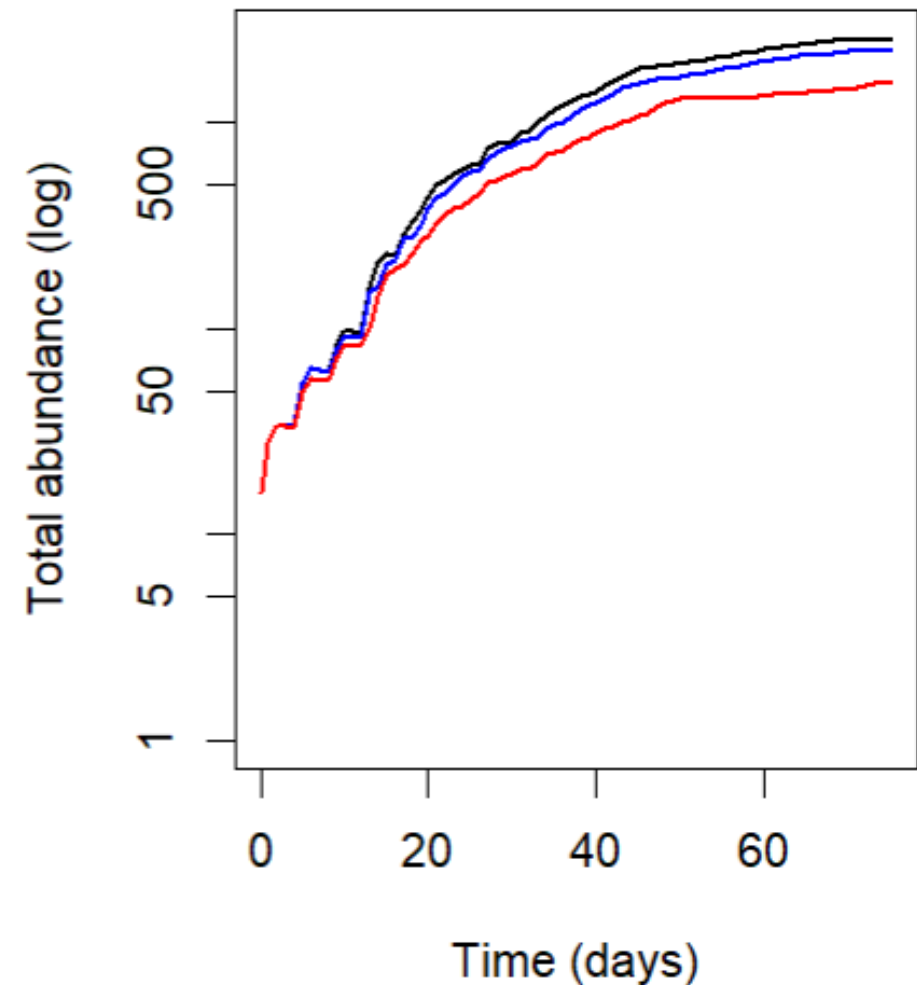
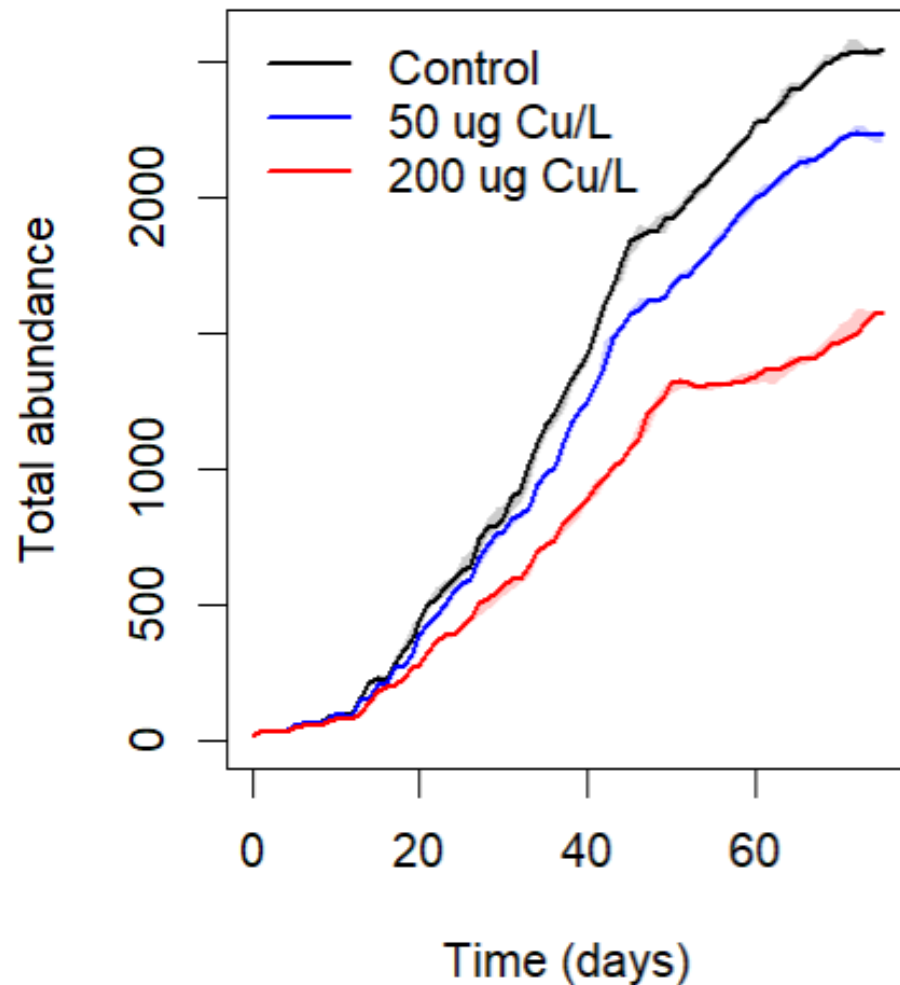
GUTs (lethal):



Exercise 3 - Coupling toxicity models to an IBM

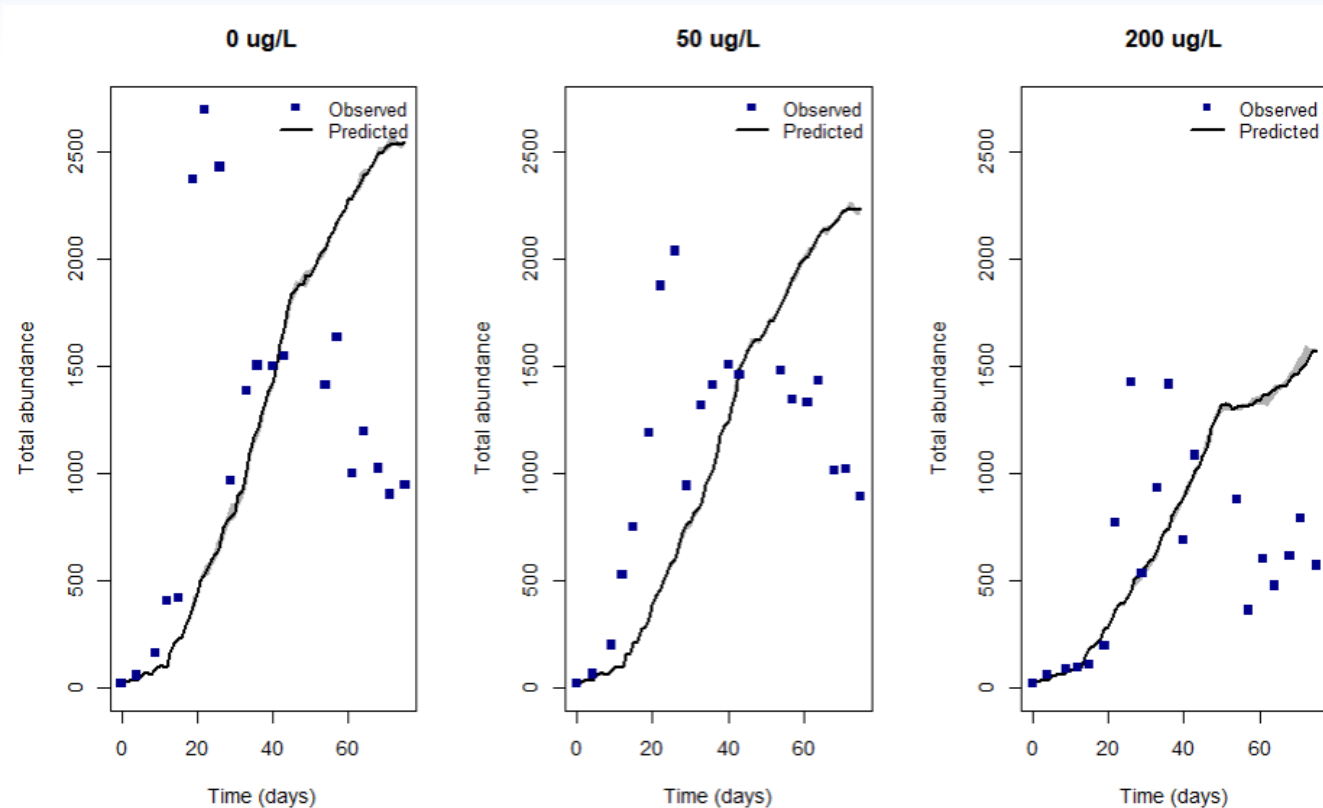
Model outputs:

Copper impacts on Daphnia



Exercise 3 - Coupling toxicity models to an IBM

Validation outputs:



Treatment
<chr>

R2_log
<dbl>

Control

0.53

50 ug Cu/L

0.51

200 ug Cu/L

0.67

Exercise 3 - Coupling toxicity models to an IBM

Predictions:

